

Sales Dataset SQL Analysis Documentation

1. Project Overview

This project analyses the Kaggle Sales Dataset using PostgreSQL. The objective was to demonstrate SQL querying, aggregation, subqueries, window functions and business analysis while answering practical business questions.

2. Tools Used

- PostgreSQL 18
- pgAdmin 4
- Kaggle Sales Dataset (CSV)

3. Database Setup

A database named sales_db was created. A table named sales was created using SQL and the CSV was imported after converting it to UTF-8 encoding. The import was verified using SELECT COUNT(*), confirming 2,823 records.

4. Core SQL Queries

Queries demonstrated SELECT, WHERE, ORDER BY, GROUP BY, HAVING, SUM, AVG and COUNT.

Key Results:

- Total Sales: \$10,032,628.85
- Average Sales per Order: \$3,553.89
- Total Orders: 2,823
- Shipped orders: 2,617 (majority of all orders)
- Highest revenue product line: Classic Cars (\$3,919,615.66)
- USA generated the highest country revenue.

5. Advanced SQL Concepts

Subquery 1 identified customers whose total spending exceeded the average customer spending.

Subquery 2 identified product lines generating above-average revenue.

ROW_NUMBER assigned sequential numbers to transactions.

RANK ranked customers by revenue.

PARTITION BY ranked customers within each country.

6. Business Problem Solving

Top Customers:

Euro Shopping Channel generated the highest revenue (\$912,294.11), followed by Mini Gifts Distributors Ltd.

Top Product Lines:

Classic Cars was the best-performing product line followed by Vintage Cars.

Revenue Trend:

Revenue increased through the year with a strong rise toward the final quarter, indicating seasonal demand.

Customer Purchasing Behaviour:

Euro Shopping Channel placed 26 distinct orders, making it both the highest spending and most active customer.

7. Query Optimization

Indexes were created on CUSTOMERNAME, PRODUCTLINE and ORDERDATE to improve query performance for customer searches, product analysis and time-based reporting. Indexing reduces the amount of data PostgreSQL scans when executing queries.

8. Conclusion

The project demonstrated the use of SQL to cleanly retrieve, summarize and analyse business data. The analysis identified high-value customers, top-performing product categories and revenue patterns that can support business decision-making.